

Week 5 - Feasibility Memo

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Final Verdict on Dataset

[!NOTE] The data is **suitable** for the development and evaluation of a baseline AI-triage model as long as real-world verification against Caribbean and paediatric cohorts is done **before** deployment.

Data Summary

The dataset contains the triage data associated with 55,121 emergency department encounters within the United States. It shows Emergency Severity Index (ESI) along with 225 other features; 200 of the features relate to the presence or absence of specific chief complaints from a patient while the other 25 relate to other factors such as vital signs. Preliminary data exploration identified 29 obviously clinically impossible values within the dataset, 25 of which belonged to the glucose column and 4 which belonged to respiratory rate as seen in *Figure 7*.

Major Feature Summary

- **225** Features relating to ESI.
- Only **29** clinically impossible values, observed in roughly **0.05%** of all cases.
- ESI level 1 and 5 only make up roughly **0.13%** and **2%** of the total cases respectively as seen in *Figure 2* and *Figure 3*.

Top 3 Concerns

The dataset has three (3) major concerns that affect its applicability in the context of Caribbean Triage:

1. No Minors in Dataset

- **Risk:** The dataset has no minors (people under the age of 18) as seen in *Figure 2*.
- **Impact:** The model has no basis to accurately evaluate paediatric trends. This would make the risk that the model is unsafe to use with children unacceptably high.
- **Mitigation:** Any models trained on the dataset must be externally validated on a paediatric sample before deployment.

2. America-centric Demographics

- **Risk:** The dataset contains a sample of the American population.
- **Impact:** The observed demographic distribution does not reflect that of most Caribbean islands. As such the disease patterns and patient presentation present in the dataset may not transfer directly to the Caribbean.
- **Mitigation:** Any resulting models trained on the dataset must be externally validated on a representative dataset from a Caribbean emergency department before being deployed in the region.

3. Class Imbalance in Acuity Level

- **Risk:** The dataset has limited examples of ESI level 1 and 5 as seen in *Figure 2* and *Figure 3*.
- **Impact:** This may cause any model trained on the dataset to be biased toward the more common ESI level 3. This may lead to poor sensitivity for critical cases, possibly increasing the rate of mistriage.
- **Mitigation:** Specific sampling techniques can be employed to mitigate the class imbalance. For example, smaller classes can be sampled at a higher rate.

Top 3 Reasons to Proceed

That said, the dataset still has many qualities that make it a valuable training tool.

1. Patient Chief Complaints Exposed

The dataset shows the chief complaints of patients. Many datasets only show vitals thus limiting the features that can be used in ESI prediction. Chief complaints reflect information directly used by triage nurses to determine acuity level.

2. No Missing Values

The dataset has no missing values (*Figure 1*), meaning that all 55,121 records can be used to train and test the model with minimal required preprocessing.

3. Most Values are Clinically Plausible

The dataset has the advantage of minimal data entry errors; most values are within the realm of plausibility as seen in as seen in *Figure 7*. This suggests the dataset is of generally good quality.

Caveats

Therefore the dataset can be used provided the following five caveats are respected:

1. Data must be validated on children

The dataset cannot be used to train a machine learning model that can reliably triage children. Paediatric data must be sourced separately provided that legal and ethical boundaries are respected.

2. Data may not generalise to Caribbean

The dataset is not centred on the Caribbean; so, local patterns may be missed.

3. Data Leakage is possible

The disposition columns are included within the dataset and must be handled carefully to prevent data leakage. This is where the model obtains data it should not have and achieves artificially inflated performance.

4. No Strong individual relationships to ESI

Preliminary analysis showed that no single feature had a strong linear relationship with ESI levels. This suggests that ESI prediction may depend more on complex multi-variable relationships than any one feature.

5. ESI Class Imbalance

Low number of ESI levels 1 and 5 cases may reduce model performance on these under-represented classes, increasing the likelihood of mistriage. Therefore the rate of false negatives for ESI level 1 cases must be tracked closely.

Top 10 Features

Rank	Feature Name	Justification
1	triage_vital_o2	The level of oxygen saturation key metric of bodily health. Low saturation is strongly correlated with more critical ESI levels.
2	cc_chestpain	Chest pain is a symptom of urgent medical emergencies like heart attacks or pulmonary embolism.
3	cc_shortnessofbreath	Shortness of breath typically implies compromised respiration which can lead to starvation of oxygen to bodily tissues.
4	cc_suicidal	Suicidal presentations have high risk of self-harm and need immediate psychiatric evaluation.
5	cc_alteredmentalstatus	Altered mental status indicates neurological, infectious and/or metabolic issues.
6	cc_motorvehiclecrash	Motor vehicle crashes are associated with potentially severe bodily trauma which must be assessed immediately.
7	cc_weakness	Weakness is may indicate medical emergencies such as stroke.
8	triage_vital_rr	High or low respiratory rate may indicate respiratory distress. Furthermore, it is also easy to acquire quickly at triage.
9	triage_vital_hr	High or low heart rates imply a cardiovascular instability and compensation for acute physiological stress.
10	cc_dentalpain	Dental pain is typically a sign of infection which can quickly spread across the body. It was shown to be correlated with ESI levels

List of Figures

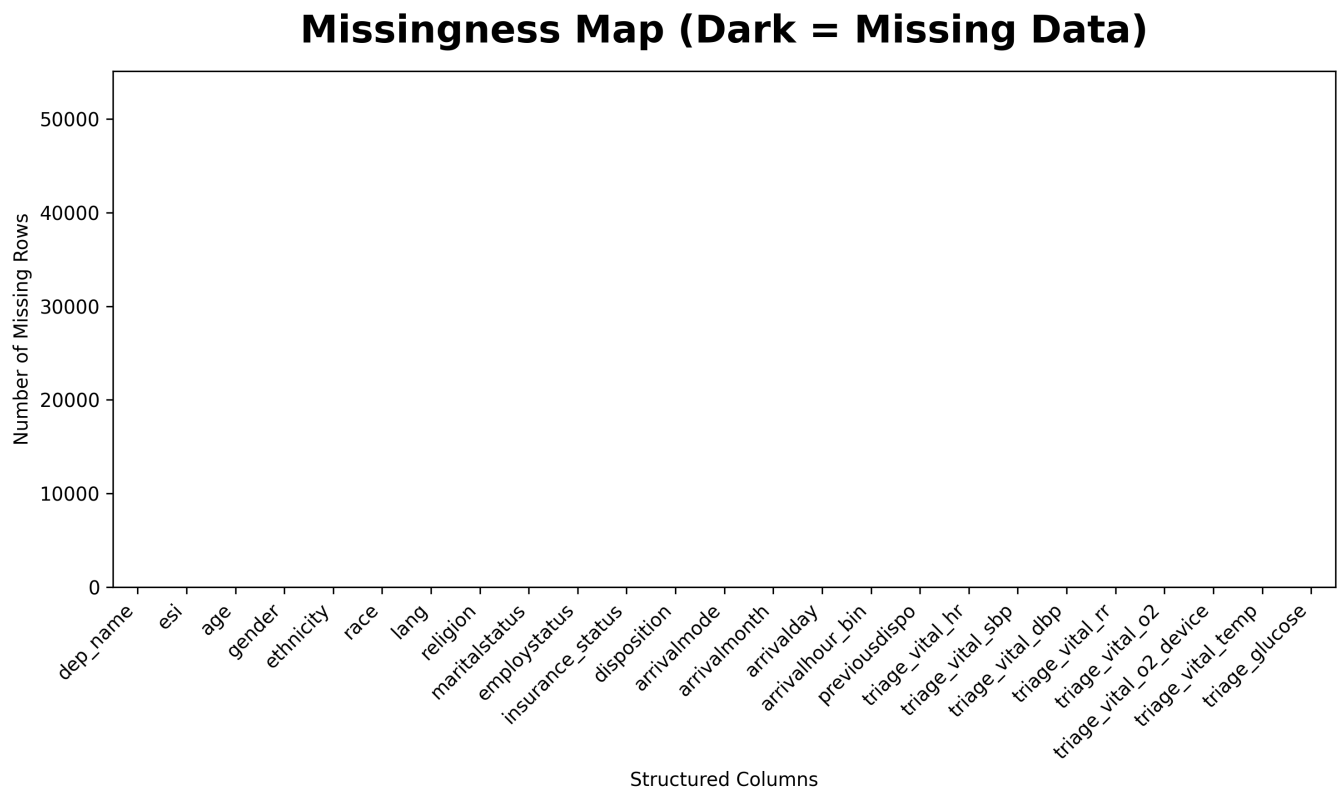


Figure 1: Heat map showing the number of missing values for the demographic features and clinical metrics (0 in this case).

Demographic Distributions

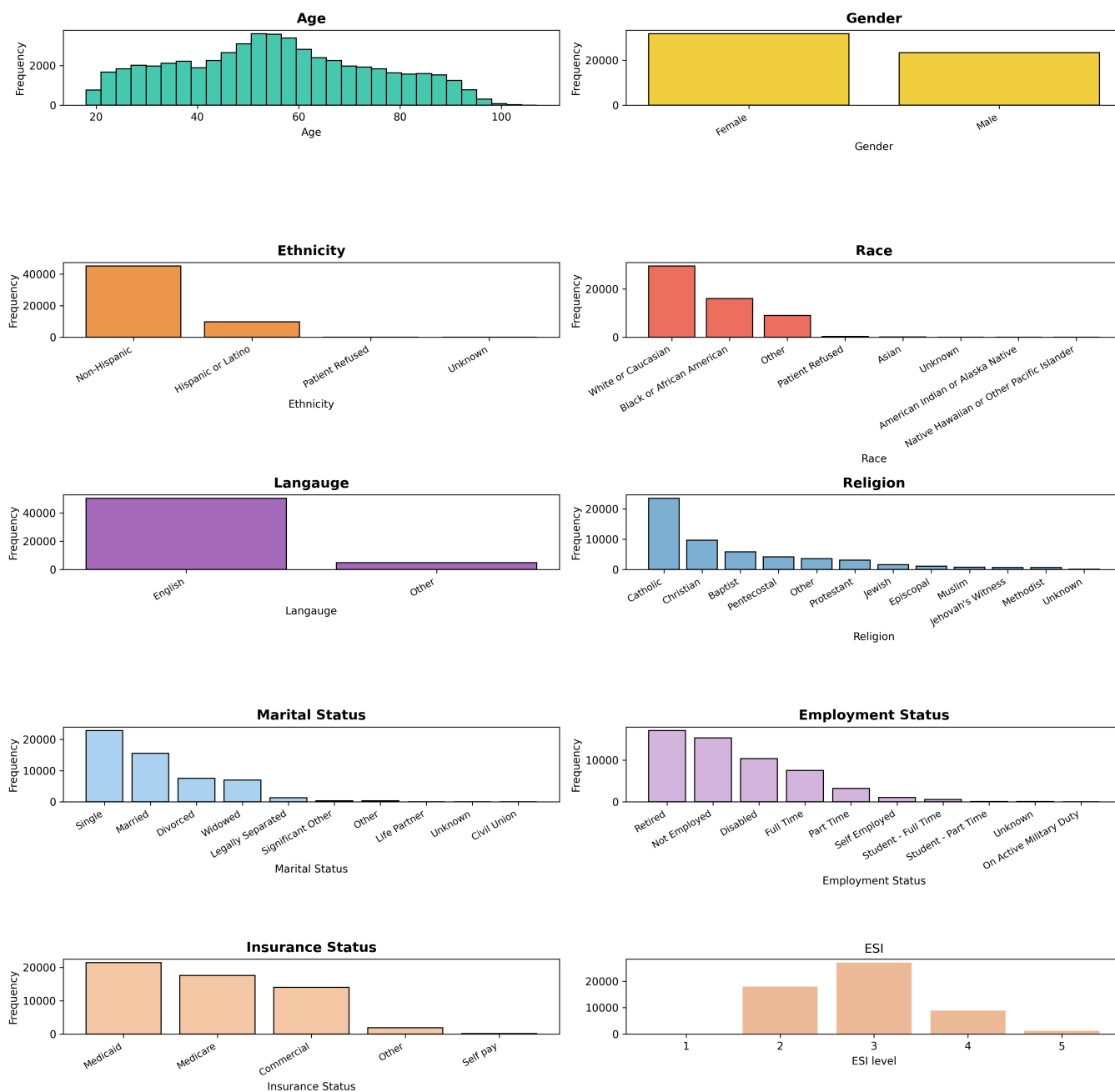


Figure 2: Bar charts showing the distribution of various demographic features in the dataset.

Vital Sign Distributions

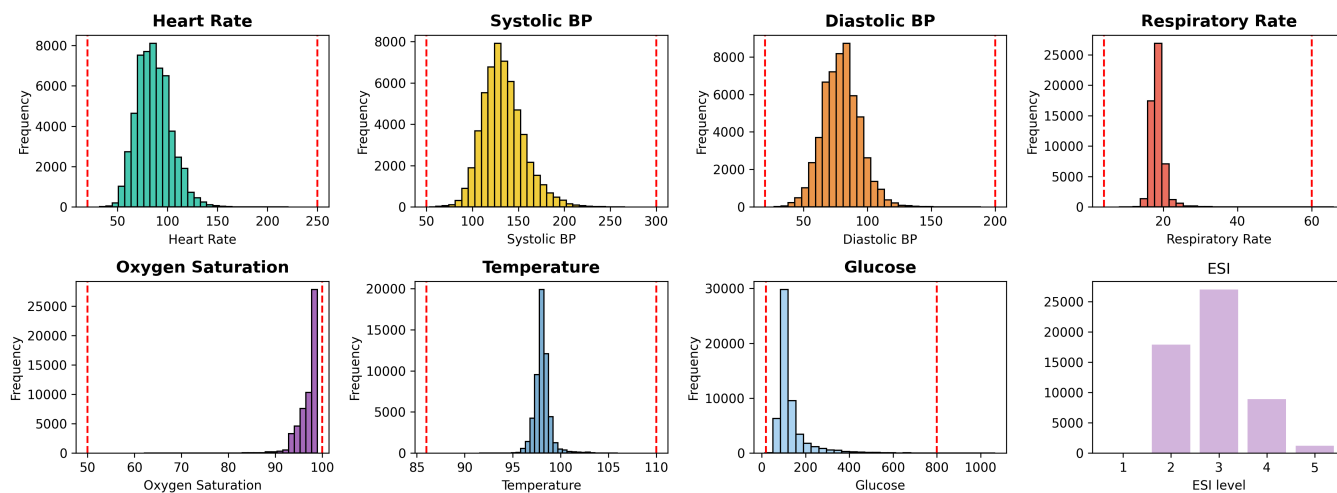


Figure 3: Bar charts showing the Distribution of various Vital Signs in the dataset.

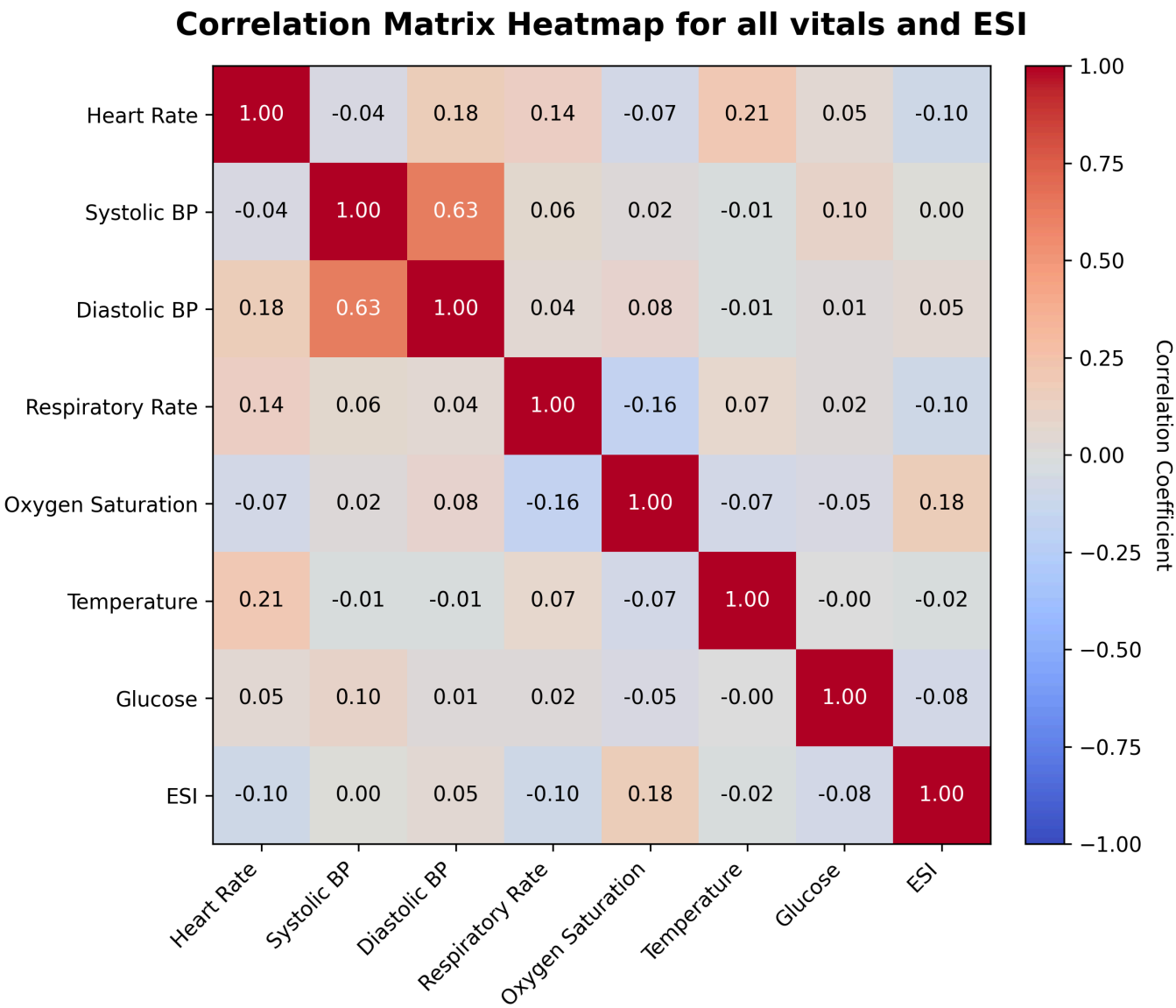


Figure 4: Correlation Matrix comparing Vital Sign and ESI within the dataset.

Correlation Matrix Heatmap for all Chief Complaint families and ESI

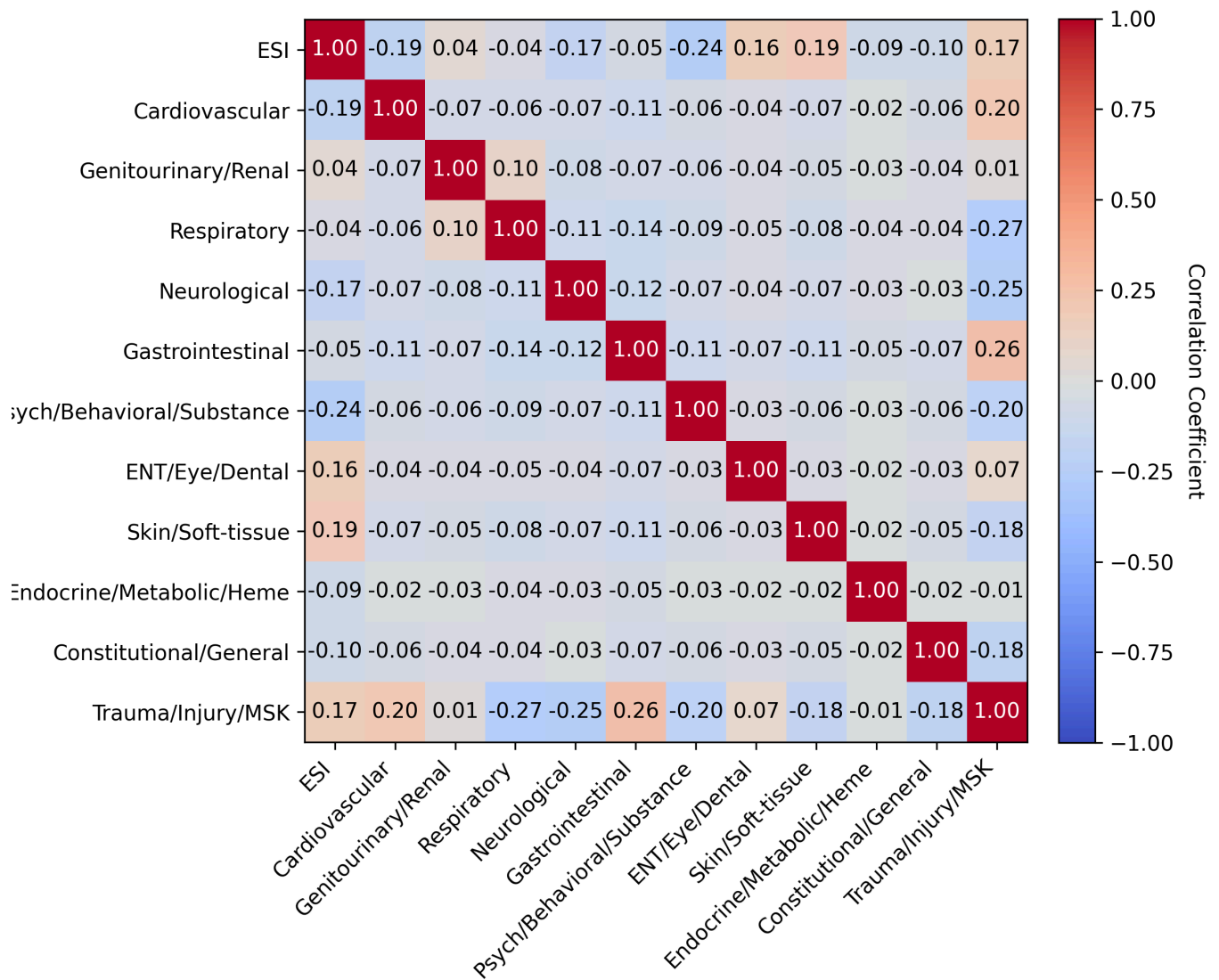


Figure 5: Correlation Matrix comparing Chief Complaints and ESI within the dataset.

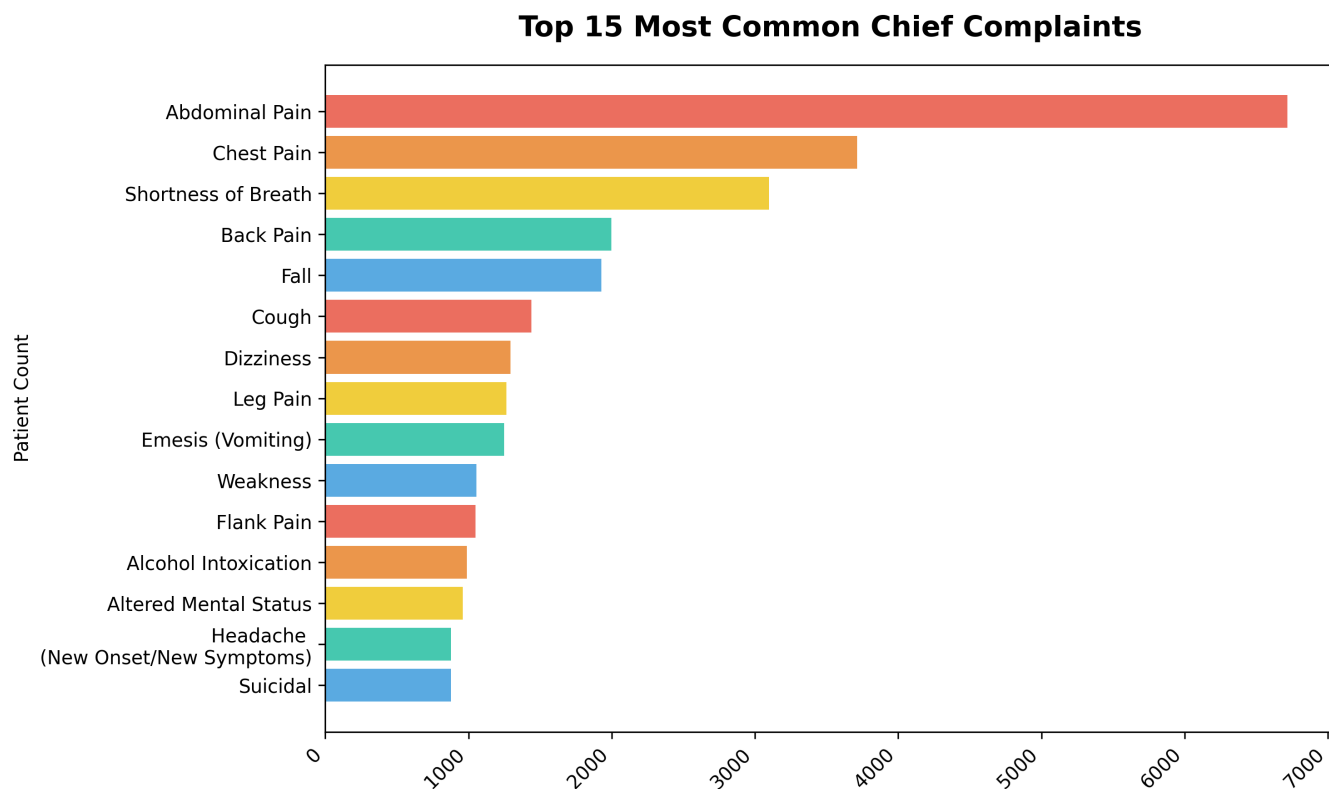


Figure 6: Horizontal Bar chart showing the Top 15 Chief Complaints in the dataset by frequency.

Frequency of Implausible vs. Abnormal for Patient Metrics

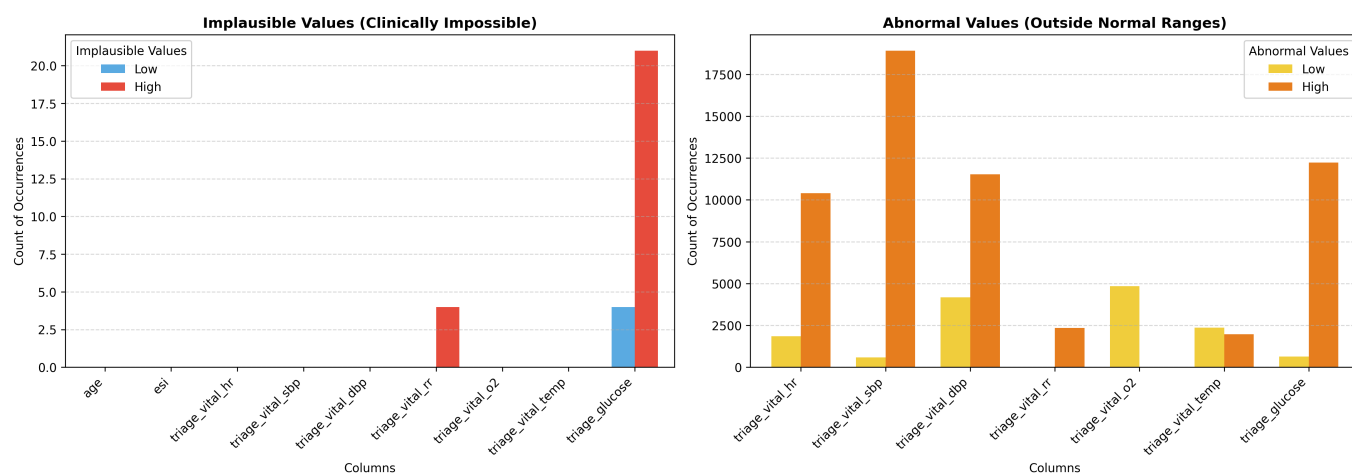


Figure 7: Bar charts showing the frequency of Implausible and Abnormal Values in the dataset.